Lamia Salsabil

Ph.D. Candidate | Multimodal LLMs, Document AI, Information Extraction

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Research Interests: Multimodal Large Language Models, Document AI, Information Extraction, Information Retrieval, Natural Language Processing, Scholarly Big Data, Digital Libraries, Web Archiving, Computational Reproducibility, and Open Science.

Education

Old Dominion University — *Norfolk, VA*
Ph.D., Computer Science — Dec. 2027 (Expected)
Advisor: Dr. Jian Wu, Candidacy passed Spring 2024 — *GPA: 3.77*

Khulna University of Engineering & Technology (KUET) — *Khulna, Bangladesh*
B.Sc., Computer Science and Engineering — Feb. 2018
CGPA: 3.35

Research Experience

Old Dominion University — *Norfolk, VA*
Graduate Research Assistant — *Jan. 2021 – Present*

Contextualized URL Extraction | *Multimodal LLMs, GPT-5-mini, Qwen3-VL* [Code]

- Created **sURL-3K**, the first benchmark for evaluating URL and context extraction, with 3,376 manually annotated URLs and their contexts from 352 scholarly papers.
- Built **VisURLX**, a multimodal LLM pipeline that reads PDF page images and extracts each URL with its context across the body, footnotes, and references, restoring footnote and reference entries into the sentences that cite them.
- Achieved **86.86% end-to-end F1**, 19–40 points above GROBID, olmOCR, and PyMuPDF pipelines.
- Presented the **first systematic comparison of image-based and text-based extraction** across accuracy, monetary cost, and computational cost, using proprietary and open-weight backends.

Dataset and Software URL Classification | *SciBERT, BertGCN, contrastive learning* [Code]

- Created **OADS-1K**, a benchmark for evaluating dataset and software URL classification, with 1,129 manually annotated URLs and their contexts from 1,574 scholarly documents.
- Developed **EnSU**, an ensemble of fine-tuned language models that assigns each URL one of six fine-grained classes from its context, crossing resource type (dataset or software) with provider (author or third party).
- Achieved **90% macro F1**, 15–22 points above few-shot GPT-4 and Claude baselines.

Computational Reproducibility Study | *DistilBERT, spaCy* [Code]

- Built a binary classifier that detects dataset and software URLs (**92% F1**) and used it on 100K ETDs to study disciplinary dependencies and chronological trends in including open datasets and software, later applied to 1.58M arXiv preprints.

Large-Scale Dataset and Software URL Repository | *VisURLX, EnSU, GROBID* Dissertation, in progress

- Building a corpus of dataset and software URLs from **16.1M** scholarly documents, using a hybrid approach combining VisURLX and GROBID for contextualized URL extraction and EnSU for classifying them.

Large-Scale ETD Collection and Metadata Quality | *NER, fuzzy matching, MySQL* [Data]

- Co-built a corpus of **533K+** ETDs from 114 universities via OAI-PMH.
- Built **ETDMIS** to validate and correct metadata with reversible version control, processing **100K** ETDs in 72 hours.
- Designed **ETD-MS v2.0**, a metadata schema with 28 entities and 160 fields.

Los Alamos National Laboratory *Los Alamos, NM*
Research Intern *Jun. 2023 – Aug. 2023*

- Built a transformer-based framework (BERT, DistilBERT, SciBERT) that classifies scholarly URLs in arXiv papers into datasets, software, media, and publishers, with **89% F1**.
- Constructed ground-truth datasets by preprocessing arXiv papers with GROBID and regular expressions.

Industry Experience

Virginia Marine Resources Commission *Fort Monroe, VA*
Software Engineer Intern *May 2024 – Aug. 2024*

- Redesigned the Freedom of Information Act (FOIA) website and digital tracking system to replace paper-based processes.
- Conducted internal training sessions on Visual Studio Code, TypeScript, Git, and MySQL for the team.
- Simulated test sales for the 2025 license portal to identify and prevent sale conflicts before launch.

Nascenia *Dhaka, Bangladesh*
Software Engineer *Jan. 2019 – Dec. 2020*

- Developed "Uddogi," a Value Added Tax (VAT) processing product for manufacturing companies with Ruby on Rails and MySQL.
- Developed a custom content management system for the UNDP-LIUPCP project using the Refinery CMS framework.
- Provisioned and deployed Nginx web servers on Linux for production traffic.

Published Work

1. **Salsabil, L.**, Obadage, R. R., Banerjee, B., Alam, S., Ingram, W. A., Fox, E. A., & Wu, J. (2026). End-to-End Contextualized URL Extraction from Scholarly Papers. *Proceedings of the 2026 ACM/IEEE Joint Conference on Digital Libraries (JCDL '26)*. https://doi.org/10.1145/3805696.3846039
2. Obadage, R. R., **Salsabil, L.**, Alam, S., Ingram, W. A., Banerjee, B., Fox, E. A., & Wu, J. (2026). URL Extraction from Scholarly Documents: A Cross-Format Comparative Analysis. *Proceedings of the 2026 ACM/IEEE Joint Conference on Digital Libraries (JCDL '26)*. https://doi.org/10.1145/3805696.3846023
3. Ingram, W. A., Peloquin, S., Dommanige, C., **Salsabil, L.**, Obadage, R. R., Banerjee, B., Alam, S., Wu, J., & Fox, E. A. (2026). A Longitudinal Panel of Open Access Data and Software URLs from PLOS Publications. *Proceedings of the 2026 ACM/IEEE Joint Conference on Digital Libraries (JCDL '26)*. https://doi.org/10.1145/3805696.3848484
4. **Salsabil, L.** (2026). Toward Preserving Open Access Datasets and Software in Scholarly Documents. *Doctoral Consortium, 2026 ACM/IEEE Joint Conference on Digital Libraries (JCDL '26)*. https://doi.org/10.1145/3805696.3847135
5. **Salsabil, L.**, Obadage, R. R., Banerjee, B., Abeysinghe, Y., Alam, S., Färber, M., Ingram, W. A., Fox, E. A., & Wu, J. (2025). Context-Based URL Classification for Open Access Datasets and Software in Scholarly Documents. *2025 ACM/IEEE Joint Conference on Digital Libraries (JCDL)*, 197–206. https://doi.org/10.1109/JCDL67857.2025.00031
6. Obadage, R. R., **Salsabil, L.**, Alam, S., Banerjee, B., Ingram, W. A., Fox, E. A., & Wu, J. (2025). Toward Robust URL Extraction for Open Science: A Study of arXiv File Formats and Temporal Trends. *Web*

Archiving and Digital Libraries Workshop (WADL@HT'25), co-located with the 36th ACM Conference on Hypertext and Social Media. https://arxiv.org/abs/2509.04759

7. Choudhury, M. H., **Salsabil, L.**, Ingram, W. A., Fox, E. A., & Wu, J. (2024). ETDPC: A Multimodality Framework for Classifying Pages in Electronic Theses and Dissertations. *Proceedings of the AAAI Conference on Artificial Intelligence*, 38(21), 22878–22884 (IAAI-24 track).
8. Ingram, W. A., Wu, J., Kahu, S. Y., Manzoor, J. A., Banerjee, B., Ahuja, A., . . . & **Salsabil, L.** (2024). Building Datasets to Support Information Extraction and Structure Parsing from Electronic Theses and Dissertations. *International Journal on Digital Libraries*, 25(2), 175–196. https://doi.org/10.1007/s00799-024-00395-4
9. **Salsabil, L.**, Wu, J., Ingram, W. A., & Fox, E. A. (2024). Toward Automatically Improving Metadata Quality of Electronic Theses and Dissertations at Scale. *2024 IEEE International Conference on Big Data (BigData)*, 8825–8827. (Poster paper)
10. **Salsabil, L.**, Wu, J., Fox, E. A., Ingram, W. A., & Pavani, A. (2024). ETD-MS v2.0: A Proposed Extended Standard for Metadata of Electronic Theses and Dissertations. *27th International Symposium on Electronic Theses and Dissertations (ETD 2024)*.
11. Choudhury, M. H., **Salsabil, L.**, Jayanetti, H. R., Wu, J., Ingram, W. A., & Fox, E. A. (2023). MetaEnhance: Metadata Quality Improvement for Electronic Theses and Dissertations of University Libraries. *2023 ACM/IEEE Joint Conference on Digital Libraries (JCDL)*, 61–65. https://doi.org/10.1109/JCDL57899.2023.00019 **[Best Short Paper Award]**
12. Escamilla, E., **Salsabil, L.**, Klein, M., Wu, J., Weigle, M. C., & Nelson, M. L. (2023). It's Not Just GitHub: Identifying Data and Software Sources Included in Publications. *International Conference on Theory and Practice of Digital Libraries (TPDL 2023)*, Springer LNCS, 195–206.
13. Wei, X., **Salsabil, L.**, & Wu, J. (2022). Theory Entity Extraction for Social and Behavioral Sciences Papers using Distant Supervision. *Proceedings of the 22nd ACM Symposium on Document Engineering (DocEng '22)*, 1–4.
14. **Salsabil, L.**, Wu, J., Choudhury, M. H., Ingram, W. A., Fox, E. A., Rajtmajer, S. M., & Giles, C. L. (2022). A Study of Computational Reproducibility using URLs Linking to Open Access Datasets and Software. *Companion Proceedings of the Web Conference 2022 (WWW '22 Companion)*, 784–788. https://doi.org/10.1145/3487553.3524658

Talks and Presentations

- **JCDL 2026**, *End-to-End Contextualized URL Extraction from Scholarly Papers*
- **JCDL 2025**, *Context-Based URL Classification for Open Access Datasets and Software in Scholarly Documents*
- **IEEE BigData 2024**, *Toward Automatically Improving Metadata Quality of Electronic Theses and Dissertations at Scale*
- **ETD 2024**, *ETD-MS v2.0: A Proposed Extended Standard for Metadata of Electronic Theses and Dissertations*
- **Sci-K Workshop at The Web Conference 2022**, *A Study of Computational Reproducibility using URLs Linking to Open Access Datasets and Software*

Honors and Awards

- **NSF Travel Grant**, 2026 ACM/IEEE Joint Conference on Digital Libraries (JCDL 2026)
- **Best Short Paper Award**, 2023 ACM/IEEE Joint Conference on Digital Libraries (JCDL 2023), for *MetaEnhance: Metadata Quality Improvement for Electronic Theses and Dissertations of University Libraries*.

Professional Service (Peer Review)

- **Reviewer**, 2026 ACM/IEEE Joint Conference on Digital Libraries (JCDL 2026), four papers
- **Reviewer**, 29th International Conference on Theory and Practice of Digital Libraries (TPDL 2025), full paper
- **Reviewer**, Joint Workshop on Extraction and Evaluation of Knowledge Entities from Scientific Documents and AI + Informetrics (AII-EEKE 2025), short paper

Teaching and Mentoring

Graduate Teaching Assistant, Department of Computer Science, Old Dominion University, Norfolk, VA

CS 120G: Introduction to Information Literacy and Research *Fall 2025*

- Assisted with lectures, recitations, and lab sessions, and prepared and graded assignments and quizzes.
- Mentored students one-on-one on research methods, literature review, and using digital libraries for information retrieval.

CS 620: Introduction to Data Science and Analytics *Fall 2024*

- Led weekly mentoring sessions on data processing, visualization, and modeling, and prepared datasets for big data exploration.
- Guided students on software tools and graded assignments and project reports.

CS 832: Human-Computer Interaction *Spring 2023*

- Guided students through HCI principles, user modeling, interface design, and empirical research projects, and graded reports and assignments.

Additional Projects

Voice-Based Housing Search Assistant | *Python, Flair, NER, REST API* [Code]

- Built **Virtual Realtor**, a voice-enabled dialogue assistant that helps users find housing that matches their requirements.

Technical Skills

Generative AI: Proprietary LLMs (GPT, Gemini, Claude), open-weight LLMs (Qwen-VL, InternVL, MiniCPM-V, Mistral), multimodal LLMs, prompt engineering

NLP & Vision Models: BERT, SciBERT, DistilBERT, SPECTER, T5, Pegasus, BertGCN, LDA2Vec, YOLO

ML Frameworks & Libraries: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, NumPy, Pandas, spaCy, Flair, BeautifulSoup, fuzzywuzzy, Weka

Document AI & OCR: GROBID, PyMuPDF, olmOCR, Amazon Textract, Tesseract

Data & Compute: PostgreSQL, MySQL, MongoDB, SQL Server, Apache Spark, Hadoop, AWS (S3), GPU computing, HPC (SLURM)

Web Development: Ruby on Rails, HTML5, CSS3, jQuery, D3.js, Leaflet.js, Nginx, Apache HTTP Server, Refinery CMS

Developer Tools: Git, Linux, Postman

Languages: Python, Java, C++, C, R, SQL, JavaScript, TypeScript, Ruby, Bash

Leadership and Voluntary Experience

- **President**, ACM-W (Association for Computing Machinery Women), Old Dominion University *2023 – 2024*
- **Student Leader**, Global Student Friendship, Norfolk, VA *2022 – 2023*
- **Mentor**, Special Group Interested in Programming Contest (SGIPC), KUET *2014 – 2018*